

Nikolai Suhhanov

Medical Simulation Developer | Unity & C# | Former General Surgeon (8+ Years) | Medical XR

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PROFESSIONAL SUMMARY

Former general surgeon with 8+ years of clinical experience transitioning into medical software development. Currently pursuing a Bachelor of Arts in Web Technologies and building medical education and surgical simulation prototypes with Unity and C#. Hands-on experience with 2D/3D interaction design, XR Interaction Toolkit, cross-platform builds, Git/GitHub, and clinically grounded training scenarios. Seeking opportunities in medical simulation, Unity development, and Medical XR.

CORE SKILLS

Development: Unity, C#, XR Interaction Toolkit, Git, GitHub, Visual Studio, 2D/3D prototyping, debugging, Windows, Android, WebGL
Web / UX: HTML5, CSS3, Bootstrap, responsive layouts, basic UX/UI methods
Medical: General Surgery, Emergency Surgery, Surgical Oncology, Human Anatomy, Patient Care, Surgical Education, Medical Simulation

SELECTED PROJECTS

XR Adhesiolysis Trainer - Surgical Simulation Prototype

Unity 3D | C# | XR Interaction Toolkit | In development

- Desktop MVP completed and adapted toward XR interaction using Unity XR Interaction Toolkit.
- Implemented grasper and scissors interaction, intestinal manipulation, adhesion tension constraints, visual feedback, and cutting logic.
- Designed the scenario around a focused surgical task rather than a full operation, based on clinical experience.

GitHub: github.com/NikolaiSuhhanov/XR_Adhesiolysis_Trainer | Demo: youtu.be/r53N9CA6B0U

Appendectomy Training - Practical Simulator

Unity 3D | C# | Windows | Android

- Interactive simulator covering the initial stages of traditional open appendectomy.
- Includes instrument selection, step-based interactions, correct/incorrect feedback, anatomy identification, and basic performance flow.

GitHub: github.com/NikolaiSuhhanov/Appendectomy-training-Practical-Simulator | Demo: youtu.be/Vr1DIYafv8U

Appendectomy Training - Theoretical Module

Unity | C# | Medical Education | Windows | Android

- Educational application covering appendix anatomy, anterior abdominal wall, surgical instruments, procedural access, image-based questions, and a short test.

GitHub: github.com/NikolaiSuhhanov/Appendectomy-training-Theoretical-Module | Demo: youtu.be/MQx6ogJtKm8

Olvar: A Short Journey - 2D Platformer

Unity 2D | C# | WebGL | Windows

- Completed three-level platformer with enemy AI, boss encounters, dialogues, UI, health/time systems, animation, audio, and WebGL deployment.

GitHub: github.com/NikolaiSuhhanov/Olvar-a-short-journey | Play: nikolai-suhhanov.itch.io/olvar-a-short-journey

CLINICAL EXPERIENCE

General Surgeon / Surgical Oncologist

Public Healthcare Institutions | 2014-2022

- Eight years of clinical experience in general, emergency, and abdominal surgery.
- Performed elective and emergency surgical procedures and participated in perioperative patient management.
- Worked in multidisciplinary teams and contributed to the training of junior colleagues and medical staff.
- Clinical background now informs the anatomy, workflow, feedback, and task design used in medical simulation projects.

EDUCATION & TRAINING

Bachelor of Arts (BA), Web Technologies

Estonian Entrepreneurship University of Applied Sciences / EEK Mainor | Sep 2025 - Jun 2028 (expected)

- Current studies include software development, web technologies, object-oriented programming, UX/UI, and related coursework.

Residency in General Surgery

2013-2015

Internship in General Surgery

2012-2013

Doctor of Medicine (MD), Medicine

Saint Petersburg State Pediatric Medical University | 2006-2012

ADDITIONAL INFORMATION

Languages: Russian - Native | English - B1/B2 | Estonian - B1

Portfolio: nikolaisuhhanov.com - project demos, GitHub repositories, CV, medical credentials, references, and supporting documents.

Availability: Open to full-time, part-time, internship, remote/hybrid opportunities, and relocation.